

Deep Learning from Sparse fNIRS Data to Augment High-Density Datasets Improves Decoding Performance

Mohammad Orabe^{1,2}, Theekshana Dissayanake^{1,2}, Shakiba Moradi^{1,2}, Alexander von Lühmann^{1,2}¹BIFOLD — Berlin Institute for the Foundations of Learning and Data, Berlin, Germany²Intelligent Biomedical Sensing (IBS) Lab, Technical University of Berlin, Berlin, Germany

Scan to view

1 Motivation

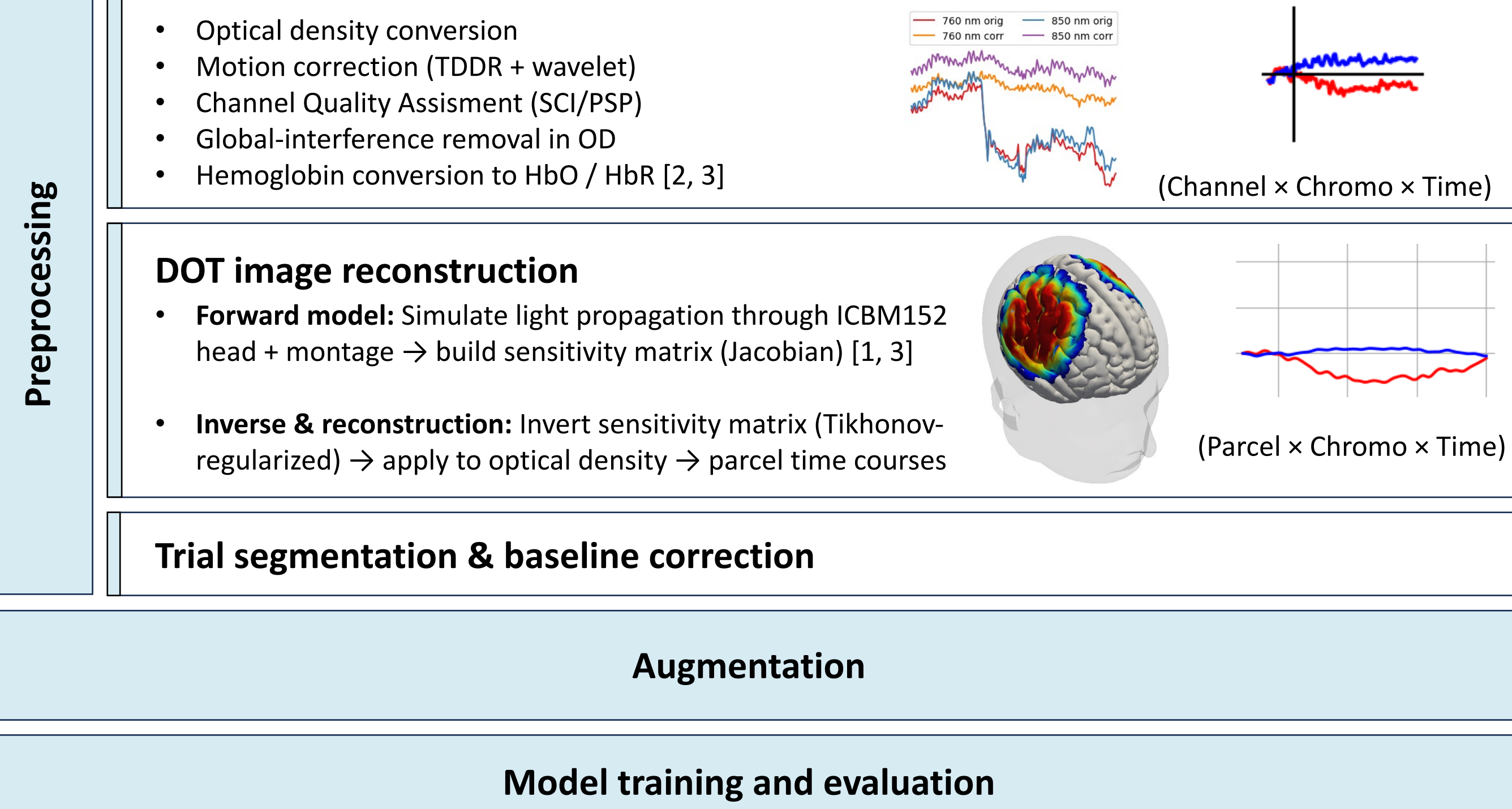
High-density functional near-infrared spectroscopy (HD-fNIRS) and diffuse optical tomography (DOT) enable spatially resolved imaging of cortical activity [1]. However, **limited dataset sizes** and substantial variability across **subjects**, **measurement configurations**, and **signal representations** challenge the generalization of deep learning models. **Data augmentation** may address these limitations, but its effectiveness has not been systematically evaluated for **HD-fNIRS** and **DOT-based classification**.

We systematically evaluate **temporal, spectral, spatial, preprocessing, reconstruction, and channel-density augmentations** across multiple datasets and both **channel- and parcel-space representations**, aiming to identify strategies that improve **subject-independent classification**.

2 Datasets

DATASET	MONTAGE	# CHANNELS	# SUBJECTS
Whole-head ball squeezing	HD	1220	13
Motor-cortex ball squeezing	HD	100	10
PFC Stroop-Rest	HD	214	16
PFC Stroop-Rest	Sparse	52	16

3 Pipeline overview



4 Augmentation Methods

1 / Channel-density sampling

Mixes 5 sparse channel subsets into dense data with different ratios.

Sparse data source **Sparse subset ratio**

same dataset other dataset 20 % 40 % 60 % 80 % 100 %

2 / EEG-inspired augmentation

Mixes 5 sparse channel subsets into dense data with 0.5 prop.

Spatial domain **Temporal domain** **Spectral domain**

ch. symmetry ch. dropout time mask gaussian noise FT surrogate frequency shift
ch. shuffle time reverse sign flip bandstop filter

3 / Time-shift

Shift the trial window in time (Temporal offset).

5 Channel-density sampling

A closer look — one of the three augmentation methods from section 04, examined here in depth

Simulate lower-density montages by removing channels first, then recomputing parcel-space signals from the sparse configuration. Each dataset yields five sparse configurations, which are then mixed back into training at fixed ratios. Dense datasets are augmented with sparse ones, never the reverse.

How a sparse configuration is built

- 01 Start from raw recordings for each subject
- 02 Filter channels by source–detector distance
- 03 Keep only valid channels within the selected distance range
- 04 Project the remaining channels to parcel space using the forward model
- 05 Restrict the parcellation to the dataset-specific valid parcels

A / Same dataset

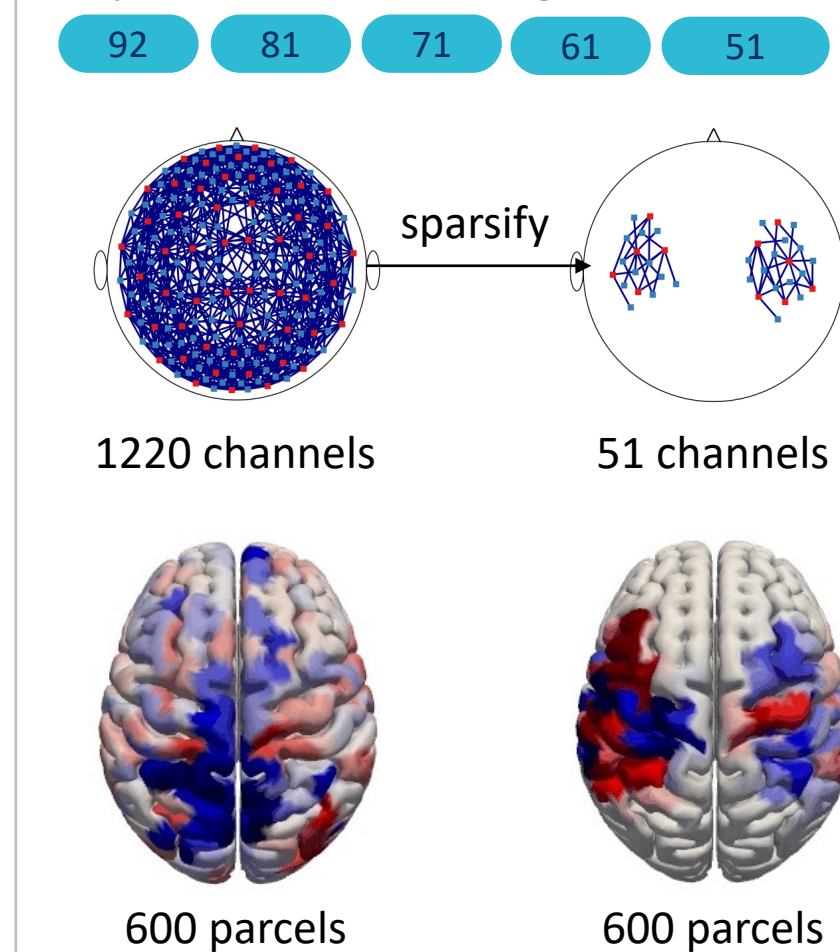
The dense data being trained on is augmented with sparse subsets drawn from that same dataset. Held-out subjects are excluded from all views.

B / different dataset

The dense data being trained on is augmented with sparse subsets drawn from a different dataset, used only for augmentation; evaluation always stays on the dense dataset.

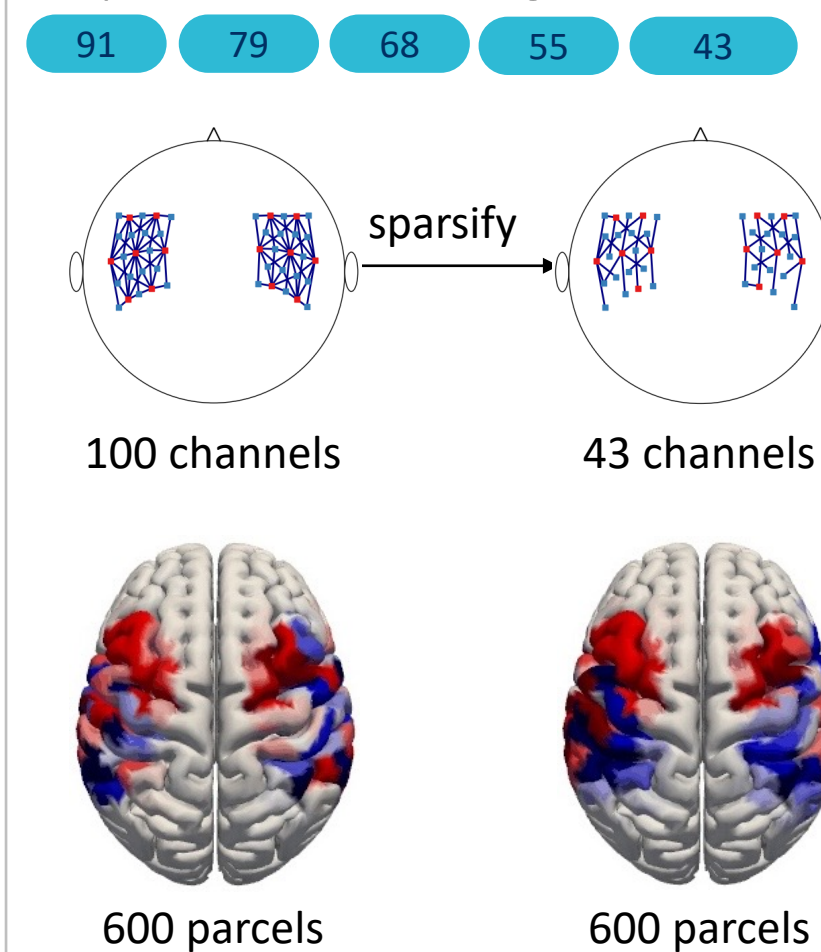
Whole-head ball squeezing

5 sparse channel configs



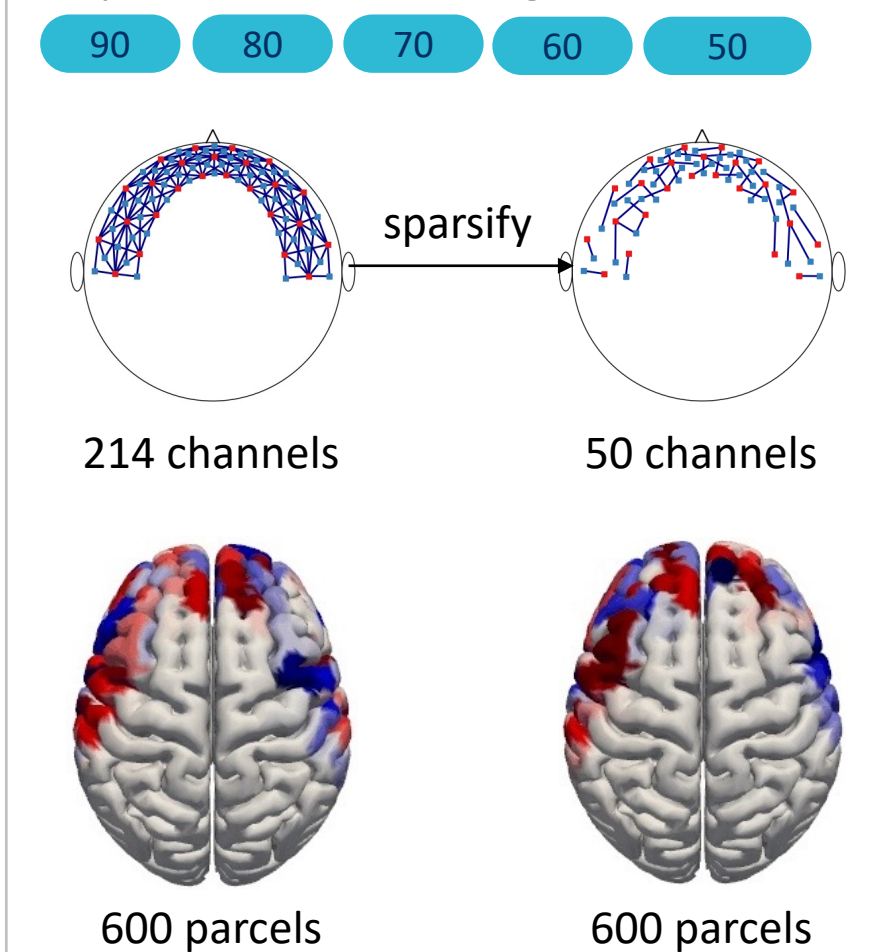
Motor-cortex ball squeezing

5 sparse channel configs

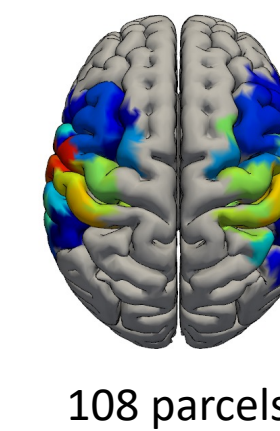


PFC Stroop-Rest

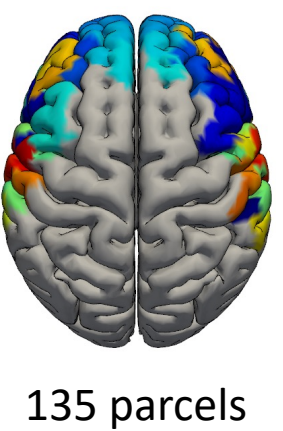
5 sparse channel configs



Motor task
Restrict to common sensitive parcels

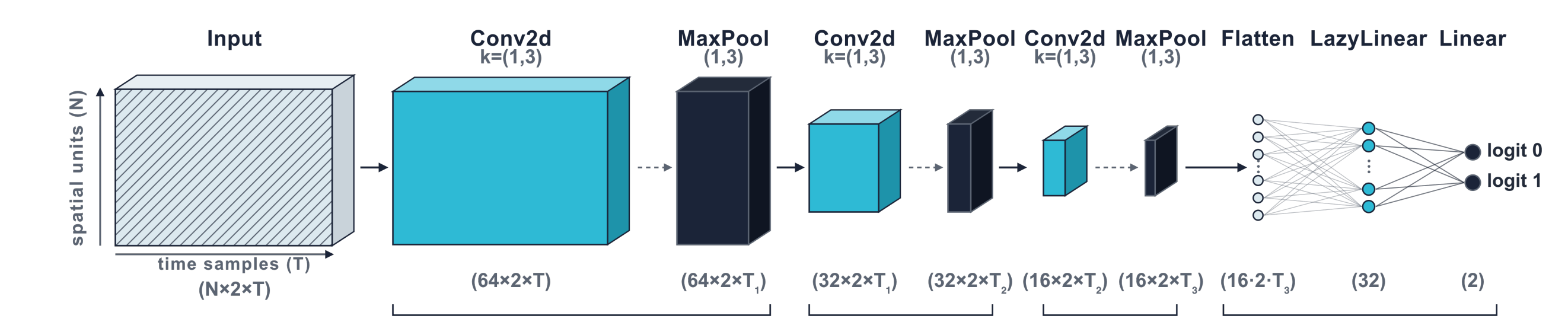


Cognitive task
Restrict to common sensitive parcels



6 Model: Convolutional Neural Network

A 2D CNN over fNIRS trial segments [4], binary classification. Spatial units N enter as the Conv2d channel dimension. Kernels and pools are (1,3): they act on time only, and the signal axis stays at 2 throughout — 2 wavelengths in channel space, 2 chromophores in parcel space.



Training & Evaluation protocol

EVALUATION

Strict leave-one-subject-out: one model per subject, tested only on that held-out subject.

AUGMENTATION

Train folds only — the held-out subject's segments stay original and un-augmented.

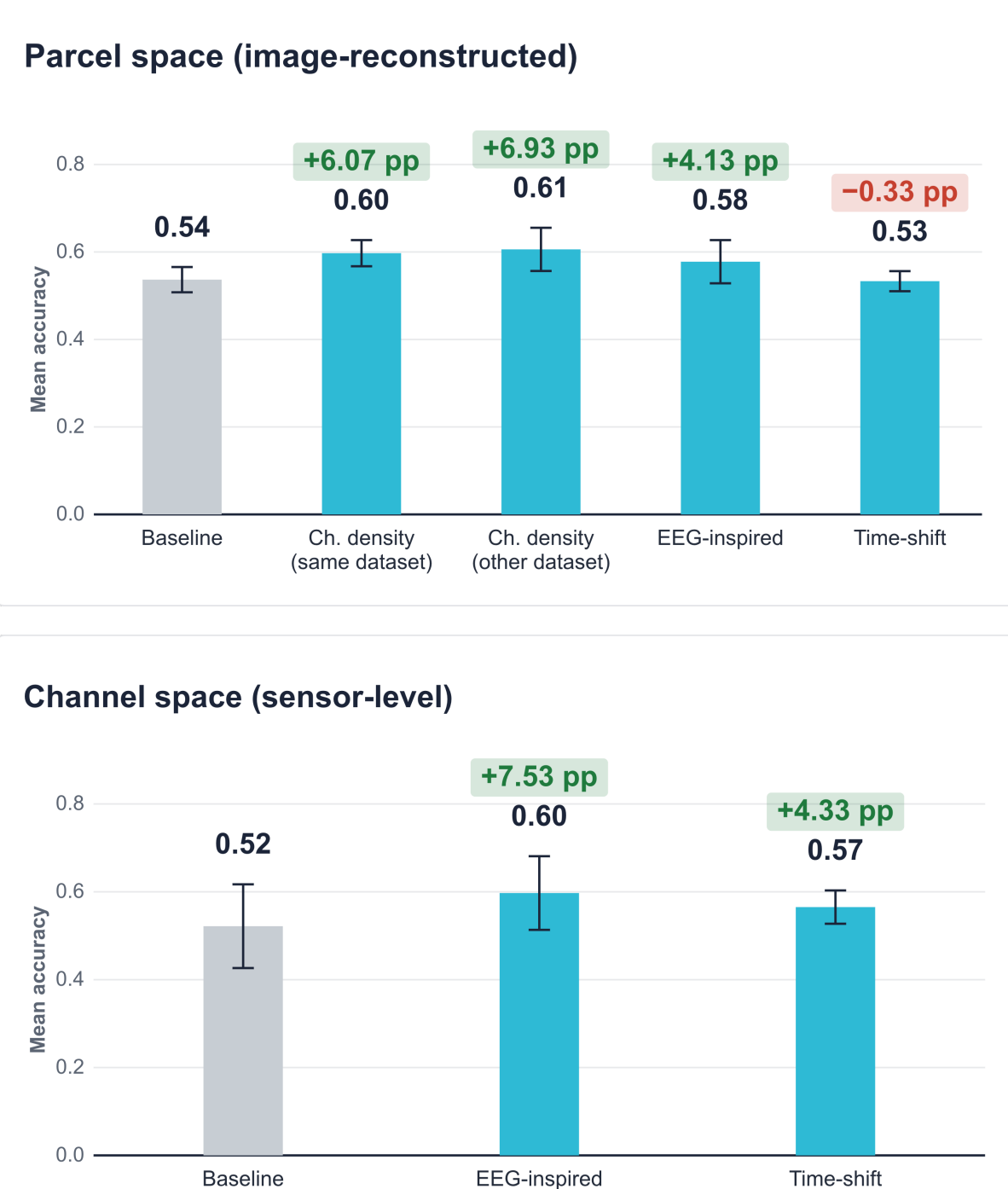
OPTIMIZATION

AdamW, 200 fixed epochs per fold, cross-entropy loss function, weight decay as regularization.

7 Results

Does data augmentation help the subjects who need it most?

Results for the worst 50% of subjects based on baseline accuracy.



Mean accuracy of the three datasets, each at its best ratio or method; deltas are absolute percentage points (pp).

	Baseline (no aug.)	Ch. density same dataset	Ch. density other dataset	EEG-inspired	Time-shift
parcel space					
Whole-head ball squeezing	0.52	0.57 (+5.1)	0.55 (+3.2)	0.52 (+0.8)	0.52 (+0.8)
Motor-cortex ball squeezing	0.57	0.63 (+5.4)	0.64 (+6.5)	0.61 (+4.0)	0.56 (-1.4)
PFC Stroop-Rest	0.52	0.60 (+7.7)	0.63 (+11.1)	0.60 (+7.6)	0.52 (-0.4)
channel space					
Whole-head ball squeezing	0.43	—	—	0.50 (+6.9)	0.54 (+10.3)
Motor-cortex ball squeezing	0.62	—	—	0.65 (+2.3)	0.61 (-1.5)
PFC Stroop-Rest	0.51	—	—	0.64 (+13.4)	0.55 (+4.2)

Mean test accuracy across the worst-performing 50% of subjects. Deltas indicate the absolute change in percentage points relative to each dataset's baseline.

8 Conclusion

01 / LIFT

Data augmentation particularly benefits the worst-performing subjects, raising accuracy in nearly every dataset/space combination tested — up to **+7.53 pp** (EEG-inspired, channel space) and **+6.93 pp** (channel-density, other dataset, parcel space).

02 / DIVERSITY

Channel-density mixing with other datasets outperforms same-dataset mixing in **2 of 3 datasets**, suggesting the gain comes from montage diversity, not just more training samples.

03 / CONSISTENCY

EEG-inspired and **channel-density** augmentation help consistently across every configuration tested; time-shift is the only strategy that sometimes hurts accuracy, despite one large win (**+10.30 pp**).

9 References

- [1] Wheelock MD, Culver JP, Eggebrecht AT. High-density diffuse optical tomography for imaging human brain function. Rev Sci Instrum. 2019. doi:10.1063/1.5086809.
- [2] Yücel MA, Lühmann AV, Scholkmann F, et al. Best practices for fNIRS publications. Neurophotonics. 2021. doi:10.1117/1.NPh.8.1.012101.
- [3] Middell E, Carlton LB, Moradi S, et al. Cedalion tutorial. Neurophotonics. 2026. doi:10.1117/1.NPh.13.S3.S32602.
- [4] Dissanayake T, Müller KR, von Lühmann A. Deep learning from diffuse optical oximetry time-series. IEEE Rev Biomed Eng. 2025. doi:10.1109/RBME.2025.3617858.

Datasets

Whole-head ball squeezing: <https://doi.org/10.82901/nemar.on007420>
Motor-cortex ball squeezing: <https://doi.org/10.1117/1.NPh.10.2.025007>
PFC Stroop-Rest: <https://doi.org/10.1117/1.NPh.12.3.035010>